EXPERIMENT NO.9

Name: Niraj Kothawade
Class:D20A
Roll No: 30
Batch:B

Aim: To implement a Private Ethereum Blockchain using Geth.

Theory:
Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy **smart contracts** and **decentralized applications (DApps)**. Unlike public blockchains such as the Ethereum mainnet, a **Private Ethereum Blockchain** is restricted to specific participants, making it ideal for **development, testing, and enterprise applications** where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through **JSON-RPC, command-line, or JavaScript consoles**.

Setting up a **Private Ethereum Network** involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**

4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The geth init command initializes the blockchain using the **genesis.json** file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**
12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Note: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

Step 1: Installing Geth on Ubuntu

sudo apt update

```
ubuntu@ubuntu:~/Desktop$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:4 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease [17.5 kB]
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [1,519 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [436 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [1,303 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [233 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1,616 kB
]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [271 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe i386 Packages [599 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [852 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [163 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [602 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [293 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [1,644 k
B]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [274 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,060 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe i386 Packages [698 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [241 kB]
```

Sudo dpkg –configure -a

```
ubuntu@ubuntu:~/Desktop$ sudo dpkg --configure -a
```

sudo apt install software-properties-common

```
ubuntu@ubuntu:~/Desktop$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-software-properties software-properties-gtk ubuntu-advantage-tools
  ubuntu-pro-client
Recommended packages:
  ubuntu-pro-client-l10n
The following NEW packages will be installed:
  ubuntu-pro-client
The following packages will be upgraded:
  python3-software-properties software-properties-common software-properties-gtk
  ubuntu-advantage-tools
4 upgraded, 1 newly installed, 0 to remove and 231 not upgraded.
Need to get 322 kB of archives.
After this operation, 1,339 kB disk space will be freed.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ubuntu-advantage-tools a
ll 31.2~22.04 [10.8 kB]
```

sudo add-apt-repository -y ppa:ethereum/ethereum

```
ubuntu@ubuntu:~/Desktop$ sudo add-apt-repository -y ppa:ethereum/ethereum
Repository: 'deb https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu/ jammy main'
More info: https://launchpad.net/~ethereum/+archive/ubuntu/ethereum
Adding repository.
Found existing deb entry in /etc/apt/sources.list.d/ethereum-ubuntu-ethereum-jammy.list
Adding deb entry to /etc/apt/sources.list.d/ethereum-ubuntu-ethereum-jammy.list
Found existing deb-src entry in /etc/apt/sources.list.d/ethereum-ubuntu-ethereum-jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/ethereum-ubuntu-ethereum-jammy.lis
t
Adding key to /etc/apt/trusted.gpg.d/ethereum-ubuntu-ethereum.gpg with fingerprint 2A518C81
9BE37D2C2031944D1C52189C923F6CA9
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease
Reading package lists... Done
```

sudo apt update

```
ubuntu@ubuntu:~/Desktop$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
231 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~/Desktop$
```

sudo apt install geth

```
ubuntu@ubuntu:~/Desktop$ sudo apt install geth
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
geth is already the newest version (1.13.14+build29502+jammy).
geth set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 231 not upgraded.
```

Step 2: Check the version of Geth on the Terminal

geth version

```
ubuntu@ubuntu:~/Desktop$ geth version
Geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8561dd7fc21b631f4a34ac16627a1
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=
GOROOT=
ubuntu@ubuntu:~/Desktop$
```

3. Step - 3: Create a Private Ethereum Network

1. Create a folder named, private_ethereum_setup

```
ubuntu@ubuntu:~/Desktop$ mkdir private_ethereum_setup
```

2. Create 2 subfolders named node1 and node2 in the folder private_ethereum_setup

```
ubuntu@ubuntu:~/Desktop$ cd private_ethereum_setup/
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ mkdir node1 node2
```

3. Create 2 accounts in the folder corresponding to node1 and node2

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node1 account new
INFO [03-31|10:20:39.436] Maximum peer count                       ETH=50 total=50
INFO [03-31|10:20:39.438] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.
comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this passwor
d.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE
Path of the secret key file: node1/keystore/UTC--2024-03-31T04-50-51.743374765Z--9f9497c56b548
90fd2f693bac26c71e4736eb2be

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ S
```

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node2 account new
INFO [03-31|10:21:31.739] Maximum peer count                       ETH=50 total=50
INFO [03-31|10:21:31.741] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.
comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this passwor
d.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xfc09dce1AB0Fe679e5cC09D88B3E99de13864433
Path of the secret key file: node2/keystore/UTC--2024-03-31T04-51-35.934981574Z--fc09dce1ab0fe
679e5cc09d88b3e99de13864433

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ 
```

4. Create a genesis.json file in the folder, private_ethereum_setup
nano genesis.json

```
ubuntu@ubuntu: ~/Desktop/private_ethereum_setup
GNU nano 6.2                                genesis.json *

"config": {
  "chainId": 1234,
  "homesteadBlock": 0,
  "eip150Block": 0,
  "eip155Block": 0,
  "eip158Block": 0,
  "byzantiumBlock": 0,
  "constantinopleBlock": 0,
  "petersburgBlock": 0,
  "istanbulBlock": 0,
  "ethash": {}
},
"nonce": "0x0",
"timestamp": "0x5f5a0a92",
"extraData": "0x0000000000000000000000000000000000000000000000000000000000000000",
"gasLimit": "0x8000000",
"difficulty": "0x4000",
"mixhash": "0x0000000000000000000000000000000000000000000000000000000000000000",
"coinbase": "0x0000000000000000000000000000000000000000",
"alloc": {
  "0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE": {
    "balance": "1000000000000000000000000"
  },
  "0xfc09dce1AB0Fe679e5cC09D88B3E99de13864433": {
    "balance": "1000000000000000000000000"
  }
}
```

5. Initialize the nodes with the genesis file

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node1 init genesis.json
INFO [03-31|10:33:21.842] Maximum peer count                       ETH=50 total=50
INFO [03-31|10:33:21.843] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.
comm: no such file or directory"
INFO [03-31|10:33:21.847] Set global gas cap                       cap=50,000,000
INFO [03-31|10:33:21.850] Initializing the KZG library             backend=gokzg
INFO [03-31|10:33:21.919] Defaulting to pebble as the backing database
INFO [03-31|10:33:21.919] Allocated cache and file handles         database=/home/ubuntu/Deskt
op/private_ethereum_setup/node1/geth/chaindata cache=16.00MiB handles=16
INFO [03-31|10:33:21.942] Opened ancient database                  database=/home/ubuntu/Deskt
op/private_ethereum_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [03-31|10:33:21.943] State schema set to default              scheme=hash
INFO [03-31|10:33:21.943] Writing custom genesis block
INFO [03-31|10:33:21.945] Persisted trie from memory database      nodes=4 size=480.00B time="
609.399µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|10:33:21.953] Successfully wrote genesis state         database=chaindata hash=c63
8fa..fe1846
INFO [03-31|10:33:21.953] Defaulting to pebble as the backing database
INFO [03-31|10:33:21.953] Allocated cache and file handles         database=/home/ubuntu/Deskt
op/private_ethereum_setup/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [03-31|10:33:21.962] Opened ancient database                  database=/home/ubuntu/Deskt
op/private_ethereum_setup/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [03-31|10:33:21.962] State schema set to default              scheme=hash
INFO [03-31|10:33:21.963] Writing custom genesis block
INFO [03-31|10:33:21.964] Persisted trie from memory database      nodes=4 size=480.00B time="
811.122µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|10:33:21.969] Successfully wrote genesis state         database=lightchaindata has
h=c638fa..fe1846
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ S
```

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node2 init genesis.json
INFO [03-31|10:33:56.477] Maximum peer count                       ETH=50 total=50
INFO [03-31|10:33:56.479] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.
comm: no such file or directory"
INFO [03-31|10:33:56.485] Set global gas cap                       cap=50,000,000
INFO [03-31|10:33:56.486] Initializing the KZG library             backend=gokzg
INFO [03-31|10:33:56.571] Defaulting to pebble as the backing database
INFO [03-31|10:33:56.571] Allocated cache and file handles         database=/home/ubuntu/Deskt
op/private_ethereum_setup/node2/geth/chaindata cache=16.00MiB handles=16
INFO [03-31|10:33:56.583] Opened ancient database                  database=/home/ubuntu/Deskt
op/private_ethereum_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [03-31|10:33:56.583] State schema set to default              scheme=hash
INFO [03-31|10:33:56.583] Writing custom genesis block
INFO [03-31|10:33:56.584] Persisted trie from memory database      nodes=4 size=480.00B time="
695.266µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|10:33:56.590] Successfully wrote genesis state         database=chaindata hash=c63
8fa..fe1846
INFO [03-31|10:33:56.590] Defaulting to pebble as the backing database
INFO [03-31|10:33:56.590] Allocated cache and file handles         database=/home/ubuntu/Deskt
op/private_ethereum_setup/node2/geth/lightchaindata cache=16.00MiB handles=16
INFO [03-31|10:33:56.602] Opened ancient database                  database=/home/ubuntu/Deskt
op/private_ethereum_setup/node2/geth/lightchaindata/ancient/chain readonly=false
INFO [03-31|10:33:56.602] State schema set to default              scheme=hash
INFO [03-31|10:33:56.602] Writing custom genesis block
INFO [03-31|10:33:56.603] Persisted trie from memory database      nodes=4 size=480.00B time="
549.872µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|10:33:56.609] Successfully wrote genesis state         database=lightchaindata has
h=c638fa..fe1846
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$
```

6. For configuring the boot node

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ sudo apt-get install bootnode
[sudo] password for ubuntu:
Sorry, try again.
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bootnode is already the newest version (1.13.14+build29502+jammy).
bootnode set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 231 not upgraded.
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$
```

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ bootnode -genkey boot.key
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ bootnode -nodekey boot.key
enode://fbbb379450b32a560c1adbc01f89cd7bf7434c64d3d61140e35f01e4dc6872e079e610606a8127a161a0ed
9d7be778636b22552290f7aaabad2e8232ed823eee@127.0.0.1:0?discport=30301
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [03-31|10:35:18.634] New local node record                    seq=1,711,861,518,633 id=ba
197519c6e384da ip=<nil> udp=0 tcp=0
```

4. Step - 4: Establish a Peer-Peer Connection between the nodes along with the bootnode

1. On the first Terminal, Use boot.key to run the boot node

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ bootnode -verbosity 5 -nodekey boot.key -addr
127.0.0.1:30305
enode://fbbb379450b32a560c1adbc01f89cd7bf7434c64d3d61140e35f01e4dc6872e079e610606a8127a161a0e
9d7be778636b22552290f7aaabad2e8232ed823eee@127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [03-31|10:40:06.489] New local node record                    seq=1,711,861,806,489 id=b
197519c6e384da ip=<nil> udp=0 tcp=0
```

2. On the second Terminal, Run Node 1

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node1 --port 30306 --bootnodes enode://f7aba85ba369
923bffd3438b4c8fde6b1f02b1c23ea0aac825ed7eac38e6230e5cadcf868e73b0e28710f4c9f685ca71a86a4911461637ae9ab2bd852939b7
7f@127.0.0.1:0?discport=30305  --networkid 1234 --unlock 0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE --password "./
password.txt" --authrpc.port 8551 --mine --miner.etherbase 0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE
INFO [03-31|11:25:31.763] Maximum peer count                       ETH=50 LES=0 total=50
INFO [03-31|11:25:31.765] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file o
r directory"
INFO [03-31|11:25:31.769] Set global gas cap                       cap=50,000,000
INFO [03-31|11:25:31.771] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [03-31|11:25:31.771] Using pebble as the backing database
INFO [03-31|11:25:31.771] Allocated cache and file handles         database=/home/ubuntu/Desktop/private_ethereum_
setup/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|11:25:31.785] Opened ancient database                  database=/home/ubuntu/Desktop/private_ethereum_
setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [03-31|11:25:31.792] Disk storage enabled for ethash caches   dir=/home/ubuntu/Desktop/private_ethereum_setup
/node1/geth/ethash count=3
INFO [03-31|11:25:31.796] Disk storage enabled for ethash DAGs     dir=/home/ubuntu/.ethash count=2
INFO [03-31|11:25:31.797] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [03-31|11:25:31.797]
INFO [03-31|11:25:31.797] -------------------------------------------------------------------------------------
-----------------------------------------------------------------
INFO [03-31|11:25:31.797] Chain ID:  1234 (unknown)
INFO [03-31|11:25:31.797] Consensus: Ethash (proof-of-work)
INFO [03-31|11:25:31.797]
INFO [03-31|11:25:31.797] Pre-Merge hard forks (block based):
INFO [03-31|11:25:31.797]  - Homestead:                   #0        (https://github.com/ethereum/execution-specs/b
lob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [03-31|11:25:31.797]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/b
lob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [03-31|11:25:31.797]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/b
lob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [03-31|11:25:31.797]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/b
```

```
ubuntu@ubuntu: ~/Desktop/private_ethereum_setup        ubuntu@ubuntu: ~/Desktop/private_ethereum_setup
il>
TRACE[03-31|11:27:49.599] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:50.100] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:50.100] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:50.601] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:50.601] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.102] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.102] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.602] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.602] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.625] << PONG/v4                               id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:51.625] >> PING/v4                               id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
DEBUG[03-31|11:27:51.625] Revalidated node                         b=16 id=1af3a1e4b66753ff checks=2
TRACE[03-31|11:27:52.103] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:52.104] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:52.604] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:52.604] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:53.105] << FINDNODE/v4                           id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
TRACE[03-31|11:27:53.105] >> NEIGHBORS/v4                          id=1af3a1e4b66753ff addr=127.0.0.1:30306 err=<n
il>
```

3. On the third Terminal, Run Node 2

```
ubuntu@ubuntu:~/Desktop/private_ethereum_setup$ geth --datadir node2 --port 30307 --bootnodes enode://f7aba85ba369
923bffd3438b4c8fde6b1f02b1c23ea0aac825ed7eac38e6230e5cadcf868e73b0e28710f4c9f685ca71a86a4911461637ae9ab2bd852939b7
7f@127.0.0.1:0?discport=30305  --networkid 1234 --keystore "./node2/keystore" --unlock 0xfc09dce1AB0Fe679e5cC09D88
B3E99de13864433 --password "./password.txt" console
```

```
INFO [03-31|11:56:49.157] Maximum peer count                       ETH=50 LES=0 total=50
INFO [03-31|11:56:49.159] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file o
r directory"
INFO [03-31|11:56:49.164] Set global gas cap                       cap=50,000,000
INFO [03-31|11:56:49.167] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [03-31|11:56:49.167] Using pebble as the backing database
INFO [03-31|11:56:49.167] Allocated cache and file handles         database=/home/ubuntu/Desktop/private_ethereum_
setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|11:56:49.192] Opened ancient database                  database=/home/ubuntu/Desktop/private_ethereum_
setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [03-31|11:56:49.202] Disk storage enabled for ethash caches   dir=/home/ubuntu/Desktop/private_ethereum_setup
/node2/geth/ethash count=3
INFO [03-31|11:56:49.202] Disk storage enabled for ethash DAGs     dir=/home/ubuntu/.ethash count=2
INFO [03-31|11:56:49.203] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [03-31|11:56:49.204]
INFO [03-31|11:56:49.204] ---------------------------------------------------------------------------------------
----------------------------------------------------------------
INFO [03-31|11:56:49.204] Chain ID:  1234 (unknown)
INFO [03-31|11:56:49.205] Consensus: Ethash (proof-of-work)
INFO [03-31|11:56:49.205]
INFO [03-31|11:56:49.205] Pre-Merge hard forks (block based):
INFO [03-31|11:56:49.205]  - Homestead:                   #0        (https://github.com/ethereum/execution-specs/b
lob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [03-31|11:56:49.205]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/b
```

5. Step - 5 : Exploring the network by attaching JavaScript console to Node 1

1. Fetch network status

```
> net
{
  listening: true,
  peerCount: 1,
  version: "1234",
  getListening: function(callback),
  getPeerCount: function(callback),
  getVersion: function(callback)
}
```

2. To fetch the number of blocks mined

```
> web3.eth.blockNumber
0
> INFO [03-23|12:15:29.133] Looking for peers                        peercount=1
 tried=0 static=0
```

3. To check the balance of the accounts

```
> eth.getBalance(eth.accounts[0])
1.000322e+24
> INFO [03-31|12:09:54.727] Generating DAG in progress
```

4. To fetch the details of the latest mined block

```
> eth.getBlock("latest")
{
  difficulty: 141501,
  extraData: "0xd883010b06846765746888676f312e32302e33856c696e7578",
  gasLimit: 114681864,
  gasUsed: 0,
  hash: "0x971fd7f7cada2a9922fad1ea258680d8b47aeb76858c0e2e1c668088af95dd46",
  logsBloom: "0x000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000",
  miner: "0x9f9497c56b54890fd2f693bac26c71e4736eb2be",
  mixHash: "0x04711352f8591ce3fe869586e2d889e92a2affa5f30d66d81d2394655fac5eed",
  nonce: "0x074eaadf2f84ee20",
  number: 161,
  parentHash: "0x4bb099d8e334270cdfd4f9014a81c05fec8e5b5d690236a6c3c3de27fe945a0e",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d49347",
  size: 538,
  stateRoot: "0x3c5c06751cbcf16f9ce7a8ae4a7593bece60d735f93c9573d92c8082cb678f33",
  timestamp: 1711867051,
  totalDifficulty: 21944603,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  uncles: []
}
> 
```

5. To fetch the details of a specific block

```
> eth.getBlock(100)
{
  difficulty: 137447,
  extraData: "0xd883010b06846765746888676f312e32302e33856c696e7578",
  gasLimit: 121724529,
  gasUsed: 0,
  hash: "0x70d88d2e2b6adbc5f77a303047feab3860428d9cf767be8a1ed614a056901a57",
  logsBloom: "0x000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000",
  miner: "0x9f9497c56b54890fd2f693bac26c71e4736eb2be",
  mixHash: "0xb0dd82c44a786d1a23d3cd2e1c7951e30cafe30ed469fc1af7b7b4410ac24d1c",
  nonce: "0x284e830e78c6a90e",
  number: 100,
  parentHash: "0x4ddc65426bffb820125bb925968d6268d30c933ff67991080ffbca37a9510cbc",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d49347",
  size: 537,
  stateRoot: "0x5d295a7b8ada8718850a4676e054e6e2dea72b739a3b2e4a4233b8be9243dea8",
  timestamp: 1711866955,
  totalDifficulty: 13436860,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  uncles: []
}
> INFO [03-31|12:12:07.308] Looking for peers                        peercount=1 tried=0 static=0
INFO [03-31|12:12:17.326] Looking for peers                        peercount=1 tried=0 static=0
```

6. To check the account balance of the peer machine, provide their Public Key

```
> eth.getBalance("0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE")
1.000322e+24
> INFO [03-31|12:13:17.464] Looking for peers
>
```

7. Fetch the details of the peers in the network

```
> admin.peers
[{
    caps: ["eth/66", "eth/67", "eth/68", "snap/1"],
    enode: "enode://dce739f3f2fae54667568d84488b8d055409c862e80d78d653bf086b250ddd431fa6c6342502f9ff8f99129b5ee0
55679fabc9335a9049ec7ae1b13941bb265e@127.0.0.1:30307",
    id: "ebf13aa79411d6e216f71bee7625c7391e7a2049d3853aed09ffb5566981dc27",
    name: "Geth/v1.11.6-stable-ea9e62ca/linux-amd64/go1.20.3",
    network: {
      inbound: false,
      localAddress: "127.0.0.1:36900",
      remoteAddress: "127.0.0.1:30307",
      static: false,
      trusted: false
    },
    protocols: {
      eth: {
        version: 68
      },
      snap: {
        version: 1
      }
    }
}]
```

8. Perform Transactions between peers in the network

```
> eth.sendTransaction({from:"0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE",to:"0xfc09dce1AB0Fe679e5cC09D88B3E99de1
3864433"})
WARN [03-31|12:16:19.599] Caller gas above allowance, capping      requested=114,569,871 cap=50,000,000
INFO [03-31|12:16:19.617] Setting new local account                address=0x9F9497c56b54890fD2F693BaC26C71e4736
eb2bE
INFO [03-31|12:16:19.619] Submitted transaction                    hash=0xf2c1e10c56397e2726e94bc166405b3ff0863d
e4757b479a1b6812debe56f33f from=0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE nonce=0 recipient=0xfc09dce1AB0Fe679e
5cC09D88B3E99de13864433 value=0
"0xf2c1e10c56397e2726e94bc166405b3ff0863de4757b479a1b6812debe56f33f"
> INFO [03-31|12:16:27.814] Looking for peers                        peercount=1 tried=0 static=0
```

9. Check the balances of sender and receiver 2

```
> eth.getBalance("0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE")

1.000322e+24
```

10. To check the details of the transaction on Node 1 Terminal

```
> eth.getTransaction("0xe181e503b93f95f2e69553091fbdd8459eda0942662c68ead0c5492779cf31c7")
{
  blockHash: null,
  blockNumber: null,
  chainId: "0x4d2",
  from: "0x9f9497c56b54890fd2f693bac26c71e4736eb2be",
  gas: 21000,
  gasPrice: 1000000000,
  hash: "0xe181e503b93f95f2e69553091fbdd8459eda0942662c68ead0c5492779cf31c7",
  input: "0x",
  nonce: 4,
  r: "0xd7bb1c00265f84c46d3c4fbf9ed8228eb791fb0e0303f1d2098989bfed21ecd1",
  s: "0x676eb39f0310ac550aeed7bade8f2988b99e64fcbcbe743cf91d2d889298c6d9",
  to: "0xfc09dce1ab0fe679e5cc09d88b3e99de13864433",
  transactionIndex: null,
  type: "0x0",
  v: "0x9c7",
  value: 90009000
}
> INFO [03-31|12:19:48.198] Looking for peers                        peercount=1 tried=0 static=0
> INFO [03-31|12:19:58.215] Looking for peers                        peercount=1 tried=0 static=0
```

11. To check the contents in the Mempool - Transaction Pool

```
> txpool.content
{
  pending: {
    0x9F9497c56b54890fD2F693BaC26C71e4736eb2bE: {
      0: {
        blockHash: null,
        blockNumber: null,
        chainId: "0x4d2",
        from: "0x9f9497c56b54890fd2f693bac26c71e4736eb2be",
        gas: "0x5208",
        gasPrice: "0x3b9aca00",
        hash: "0xf2c1e10c56397e2726e94bc166405b3ff0863de4757b479a1b6812debe56f33f",
        input: "0x",
        nonce: "0x0",
        r: "0xfe80302530833e03cb38f654004a63d2b812eeab0d15ac0c743c02959e32b17e",
        s: "0x532a68167bf51184d46643bc35d828d941fc9a75a043b4b37010a7409cb41f1c",
        to: "0xfc09dce1ab0fe679e5cc09d88b3e99de13864433",
        transactionIndex: null,
        type: "0x0",
        v: "0x9c8",
        value: "0x0"
      },
      1: {
        blockHash: null,
        blockNumber: null,
        chainId: "0x4d2",
        from: "0x9f9497c56b54890fd2f693bac26c71e4736eb2be",
        gas: "0x5208",
        gasPrice: "0x3b9aca00",
        hash: "0xdde43d6b52e779b78d33a4e5c311ce24b71fc7c7434c93464d662ce20f81ec3c",
        input: "0x",
        nonce: "0x1",
        r: "0xa1abb8a85a39d5da37ac14f8275a2eb5034e8182bd4cfa7b3523d631020188f3",
        s: "0xfb96f3fb07299f86289f4a30805b0872a36b091f9c9c65e577de70adb609e11",
        to: "0xfc09dce1ab0fe679e5cc09d88b3e99de13864433",
        transactionIndex: null,
```

12. To check the status of the Mempool - Transaction Pool

```
> txpool.status
{
  pending: 5,
  queued: 0
}
> INFO [03-31|12:21:38.389] Looking for peers                        peercount=1 tried=0 static=0
```

Conclusion:-

The experiment successfully implemented a **Private Ethereum Blockchain** using **Geth** and **Clique (PoA)** consensus. By initializing a custom **genesis block**, an isolated network was established for zero-cost testing. This setup proved that authorized signers can effectively validate transactions without the computational overhead of public networks, providing a secure environment for enterprise DApp development.